

WEEK 05 | WED 06·17 | ACT II REVIEW

The Hall, *Lit.*

You have tuned the room for sound. Now you negotiate with the sun. This package is the evidence that your aperture strategy, your shading decisions, and your choice of analysis zone were driven by solar geometry and performance data – not default glazing.

ACT II ARC

Week 4 – Solar geometry and sun path; LightStanza model setup; first solstice sun studies; aperture stance committed to Decision Log. **Week 5** – Full climate-based sDA/ASE analysis; shading device sizing; glare control strategy; LightStanza lab. **Week 6** – Act II pin-up review; submit this package by Wed 06·24 at 11:59 PM.

DUE	WEIGHT
Wed 06·24 <i>11:59 PM · Canvas</i>	10% <i>20 pts raw</i>
SUBMISSION	NAAB
PDF + Text <i>two parts, one assignment</i>	SC.4B <i>+ SC.4E secondary</i>

What You Are *Submitting*.

Act II closes with a two-part Canvas submission: a **PDF package** containing your drawings, LightStanza output, shading device calculations, and pin-up slides, and a **pasted text entry** of your Decision Log entries from Weeks 4–5. Both parts are required. A missing Decision Log text entry will score zero on Criterion 5 regardless of the quality of the PDF.

The PDF is graded against a five-criterion rubric worth 20 points total. The central question for this act: *does the aperture strategy serve both the prefunction and the hall?* The prefunction must be demonstrably daylit by the numbers. The hall must be demonstrably protected from uncontrolled daylight by design intent – not by accident of orientation.

LightStanza is the required analysis tool for this act. Results must appear as exported maps or screenshots, not as verbal descriptions of what the software showed.

Export everything into a single PDF. Page order does not need to follow this list exactly, but all items must be present. Label each page or section clearly.

□ **Aperture Strategy Plan + Section**

Your current plan and at least one building section showing window placement, compass orientation, and the boundary between the hall blackout zone and the prefunction daylighting zone. Glass locations labeled with approximate window-to-wall ratio (WWR) by facade. Shading devices shown in section with dimensions. North arrow required. Minimum 1/8" = 1' scale or equivalent.

□ **LightStanza Sun Study Screenshots**

Minimum four solar position screenshots, each labeled with date, time (local solar), and camera orientation. Required: June 21 at 10:00 AM and 2:00 PM; December 21 at solar noon. Fourth image: your choice of worst-case condition for your specific orientation. Annotate each image to identify sunlit surfaces and shadow edges relevant to your aperture decision.

□ **sDA & ASE Maps – Prefunction Analysis Zone**

LightStanza climate-based analysis output for your primary prefunction floor area. Required maps: Spatial Daylight Autonomy (sDA) and Annual Sunlight Exposure (ASE) as floor plan heatmaps.

Annotate the maps with the numeric sDA% and ASE% results. Compare to LEED v4 targets: **sDA ≥ 55%** for nominally daylit; **ASE ≤ 10%** for glare compliance. If your prefunction fails either target, include a written note identifying the specific cause and a proposed design revision.

□ **Shading Device Calculation**

For your primary south or east/west aperture: calculate the required overhang projection depth using the solar altitude angles for Detroit (42.3°N). Show the calculation: noon altitude on June 21 (71°) sets the maximum shading depth; noon altitude on December 21 (25°) sets the minimum admission angle. Derive overhang depth as a ratio of window head height. Sketch or annotate the section with the resulting geometry. If your primary aperture faces north, explain why north glazing requires no horizontal shading device and specify how you addressed glare from diffuse sky luminance instead.

□ **Glare Control Strategy**

Identify the highest-risk glare condition in your prefunction (time of day, facade, sun angle). Specify the glare control approach: exterior fixed shades, fritted or diffusing glass, interior adjustable blinds, or a combination. If LightStanza includes a Daylight Glare Probability (DGP) output, include the map and note whether the result falls in the imperceptible (<0.35) or perceptible range. If DGP is not available, provide a qualitative assessment referencing luminance contrast ratios and window reveal depth.



Act II Pin-Up Slides

Your presentation from the Week 6 in-class review, exported as PDF pages. Graded on content, not slide design. A concise 6–8 slides covering: aperture strategy rationale, sDA/ASE results, shading device approach, and the acoustic–daylight trade-off you navigated (where did the daylighting strategy have to accommodate decisions already locked in by Act I).

§ 03 Decision Log *Text Entry.*

In the Canvas text entry box below the file upload, paste the full text of your Decision Log entries from Weeks 4–5 (Entries 5–8, or however many you have written through Act II close). Do not summarize or edit – paste as written. Entry 5 (aperture stance, written in Week 4 lab) is the minimum required entry for this act; entries 6–8 should document how your analysis results caused you to revise, confirm, or abandon your initial stance.

FORMAT FOR THE TEXT PASTE

Copy each entry from your Google Doc exactly as written. Separate entries with a blank line. You do not need to reformat.

Entry 5 • Week 4

Decision: South prefunction facade receives primary glazing at 40% WWR...

System: Daylighting

Space: Prefunction Lobby

Reasoning: South face at 42°N admits low-angle Dec 21 sun at 25° altitude; a 2 ft overhang blocks Jun 21 noon sun at 71° while admitting winter sun...

Trade-off: South glass conflicts with the acoustic isolation layer on the south lobby wall – will need to coordinate with the STC assembly...

Reconsider: If sDA comes in below 55%, I will increase WWR or add a light shelf...

Entry 6 • Week 5

Decision: After running sDA/ASE, revised south WWR from 40% to 32% to bring ASE below 10%...

...

Five criteria, four points each, on a five-level scale: *Exceeds* (4) · *Mastery* (3) · *Near* (2) · *Below* (1) · *No Evidence* (0). Scores map directly to the gradebook: 20 raw points = 10% of the course grade. Each criterion is evaluated independently. *Mastery* represents full competency at the graduate level for this stage of the project.

01 Aperture Strategy & *Geometric Logic*

SC.4B

4 pts

EXCEEDS 4	Plan and section clearly distinguish the hall blackout zone from the prefunction daylighting zone. Aperture placement on each facade is explicitly justified by compass orientation and solar geometry: south apertures reference the controllability of horizontal shading; north apertures reference diffuse sky consistency; east/west apertures identify the glare risk and the vertical fin or setback response. WWR is stated by facade. The strategy is coherent – glass placement on any face cannot be read as a default or arbitrary choice.
MASTERY 3	Blackout zone and daylighting zone are identified. Primary aperture facades are selected with reasonable justification. Minor gaps: one facade's glazing rationale is unclear, or WWR is not stated. Section shows glass location but shading device geometry is approximate.
NEAR 2	Glass is placed on the drawings but the plan does not distinguish the hall from the prefunction as daylighting zones. Aperture placement appears intuitive rather than derived from solar geometry or program requirements. No WWR stated.
BELOW 1	No plan or section submitted showing aperture placement. Or: glazing is placed uniformly on all facades with no differentiation between the hall and prefunction, and no reference to orientation.
NO EVIDENCE 0	No drawings submitted. Neither the aperture plan nor any daylighting/blackout zoning is present in any form.

02 Quantitative Analysis — *sDA* & *ASE*

SC.4B

SC.4E

4 pts

EXCEEDS 4	sDA and ASE maps for the prefunction analysis zone are included as LightStanza heatmap exports. Numeric sDA% and ASE% values are annotated on or adjacent to the maps. Results are compared explicitly to LEED v4 targets (sDA $\geq$ 55%, ASE $\leq$ 10%). If either target is missed, the student identifies the specific cause (orientation, WWR, insufficient shading) and proposes a revision. The analysis zone boundary matches the prefunction floor area shown on the plan – the student understands what they measured.
MASTERY 3	sDA and ASE maps are present with numeric results annotated. LEED targets are referenced. If a target is missed, it is acknowledged. Minor gap: analysis zone boundary is not shown on the plan, or the student describes results without connecting them to aperture decisions.
NEAR 2	LightStanza output is included but only as sun study screenshots – no sDA/ASE climate-based analysis maps. Or: maps are present but numeric results are not annotated and LEED targets are not referenced.
BELOW 1	No LightStanza output submitted. Or: only a description of what the analysis showed, with no maps or numeric data. The submission does not demonstrate that climate-based analysis was run.
NO EVIDENCE 0	No daylighting analysis of any kind submitted – no maps, no screenshots, no numbers.

03 Shading Device *Design*

SC.4B

SC.4E

4 pts

EXCEEDS 4	Overhang projection depth is calculated using Detroit solar altitude angles: the June 21 noon altitude (71°) determines the required shading depth; the December 21 noon altitude (25°) confirms winter sun admission. Calculation is shown with the formula (overhang depth = head height ÷ tan(summer altitude)), the numeric result, and a section sketch verifying the geometry. Device type is matched to orientation: horizontal overhang for south; vertical fins, louvers, or deep setback for east/west; explanation of why north requires no solar shading device. Device dimensions appear on the section drawing.
MASTERY 3	Shading device is specified and shown in section with approximate dimensions. Solar altitude angles for summer and winter are referenced. Calculation result is stated but derivation is incomplete or formula is not shown. Device type is appropriate to orientation.
NEAR 2	A shading device is mentioned or shown without calculation. Depth or dimensions are estimated without reference to solar geometry. Device type may not be appropriate to the orientation (e.g., horizontal overhang specified for an east facade without acknowledgment of its limitations).
BELOW 1	No shading device shown or calculated. Or: shading is addressed only verbally (“blinds will be provided”) with no geometric specification, no solar angle reference, and no section drawing showing the device.
NO EVIDENCE 0	No shading device is shown, drawn, calculated, or mentioned anywhere in the submission.

04 Glare Control & *Visual Comfort*

SC.4B

4 pts

EXCEEDS 4	The highest-risk glare condition in the prefunction is identified by time, date, and facade (e.g., “west lobby at 5 PM on June 21”). A specific glare control strategy is specified: exterior fixed shades, fritted or diffusing glass, or occupant-controlled interior shades, with reasoning for the choice. If LightStanza DGP output is available, it is included and interpreted (<0.35 = imperceptible; 0.35–0.45 = perceptible; >0.45 = disturbing). Aperture reveals or splay dimensions are noted as a contrast-reduction strategy. The control strategy is drawn or annotated on the section.
MASTERY 3	Glare risk is identified by facade. A control strategy is specified with a stated rationale. DGP output is included if available, or absence is acknowledged. Minor gap: the glare condition is described without a specific time/date reference, or the control strategy is stated without appearing on the drawings.
NEAR 2	Glare is mentioned as a concern without identifying a specific condition or specifying a control strategy. “Blinds will be provided” without discussion of type, operability, or luminance contrast management.
BELOW 1	No glare control addressed. Or: high-ASE result in the analysis is not connected to any design response. The submission does not demonstrate awareness that occupant discomfort is a design responsibility.
NO EVIDENCE 0	Glare and visual comfort are not addressed anywhere in the package.

05 Decision Log *Quality*

SC.4B

4 pts

EXCEEDS 4	Four or more entries covering Weeks 4–5. At least one entry references a specific solar geometry fact or LightStanza result (e.g., “reduced south WWR from 40% to 32% after ASE came in at 18%, above the 10% LEED threshold”). At least one entry acknowledges a trade-off with the acoustic decisions locked in during Act I (e.g., aperture placement constrained by the STC assembly, or glass area limited by the need to maintain absorption coefficient targets). Reasoning is in the student’s own voice and demonstrates that the sDA/ASE data changed or confirmed a design decision.
MASTERY 3	Three or more entries covering Weeks 4–5. At least one entry connects a decision to a solar geometry fact or analysis result. Acoustic trade-off is acknowledged on at least one entry. Trade-off field is completed on most entries.
NEAR 2	Two entries submitted, or entries describe what was done without referencing analysis results. sDA/ASE numbers do not appear in any entry. No acknowledgment of the acoustic–daylight relationship.
BELOW 1	Zero or one entry submitted, or entries are placeholder text. No connection between log content and the daylighting work visible in the PDF package.
NO EVIDENCE 0	No Decision Log text entry submitted for Act II.

This assignment addresses two Student Performance Criteria from the NAAB 2020 Conditions. Criteria 1, 4, and 5 map primarily to SC.4B; Criteria 2 and 3 draw on both.

SC . 4 B**Environmental Systems**

Principles of environmental systems design; application to problems involving daylighting, lighting systems, and acoustics; tools used for performance assessment; analysis across different geographic regions.

SC . 4 E**Building Service Systems**

Principles of building service systems; appropriate application of quantitative performance analysis; use of advanced simulation technologies to assess environmental performance.

Use these as calibration guides, not guarantees. A student who scores Mastery on all five criteria earns 15/20 (B range), which is the expected outcome for a graduate student running climate-based daylighting analysis for the first time.

SCORE	GRADE EQUIV.	WHAT THE WORK SHOWS
18 – 20	A	Strong integration of analysis and design intent across all five criteria. The sDA/ASE result visibly changed something in the design – a revised WWR, an added shading device, a relocated aperture. Decision Log shows genuine iteration between data and decision. The acoustic–daylight trade-off is named and resolved, not ignored.
14 – 17	B	Solid technical work with a minor gap in one criterion. Analysis is present and results are annotated. Shading device is specified with approximate geometry. Log is substantive but may lack a cross-system trade-off entry.
10 – 13	C	Analysis maps are present but numeric results are not referenced in the design narrative. Shading device is shown without calculation. Aperture strategy reads as intuitive rather than derived. Log entries describe what was done rather than why numbers changed the approach.
6 – 9	D	Major gaps in technical content – sun studies only (no sDA/ASE), no shading device calculation, or no plan distinguishing hall from prefunction. Decision Log missing or one entry only. The PDF does not demonstrate that solar geometry or analysis data drove any design decision.
≤ 5	F	Did not meaningfully engage with the daylighting analysis problem. No LightStanza output present. Drawings unchanged from Week 4. Required elements absent. Contact the instructor immediately if you are in this range before the deadline.